AMPLENESS OF A TWIST OF A PULLBACK ALONG A COMPOSITE, FOR A QUASI-COMPACT BASE

THE STACKS PROJECT AUTHORS

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Editorial note. Six-line proof body; the bookkeeping it does (choosing a finite affine covering from the quasi-compactness hypothesis, then invoking the affine criterion for relative ampleness and the twist lemma on the pieces) is the actual content of the lemma, so the span is a proof and not a pointer. The quasi-compactness hypothesis is exactly what makes the covering finite.

Allowed standard prerequisites (editorial).

- The affine open criterion for relative ampleness. Lemma \ref{lemma-characterize-relatively-ample}, cited at proof lines 8900-8902. Reduces the assertion to the pieces of a finite affine covering.
- Ampleness of a twist of a pullback along a morphism into an ample situation. Lemma \ref{lemma-pullback-ample-tensor-relatively-ample}, cited at proof lines 8904-8906. Gives ampleness on each affine piece, hence for the composite.
- Quasi-compactness of the base, so that a finite affine covering exists. the hypothesis of the statement, used at proof line 8899. Supplies the finite covering the proof works with.

Source reference. Original lemma 0C4K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15230–15230. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2448–2448. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0892: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10482–10482. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $g : Y \to S$ and $f : X \to Y$ be morphisms of schemes. Let $\mathcal{M}$ be an invertible $\mathcal{O}_Y$-module. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. If S is quasi-compact, $\mathcal{M}$ is g-ample, and $\mathcal{L}$ is f-ample, then $\mathcal{L} \otimes f^*\mathcal{M}^{\otimes a}$ is $g \circ f$-ample for $a \gg 0$.*

Proof. Let $S = \bigcup_{i=1,\ldots,n} V_i$ be a finite affine open covering. By Lemma 01VJ it suffices to prove that $\mathcal{L} \otimes f^*\mathcal{M}^{\otimes a}$ is ample on $(g \circ f)^{-1}(V_i)$ for $i = 1, \ldots, n$. Thus the lemma follows from Lemma 0892. □